

PayStream Glossary

Payment Methods

API_Direct

Merchants integrate directly with PayStream's REST API to process payments. Highest transaction volume (60%) and lowest fee (1.1%). Used primarily by tech-savvy merchants and SMEs with development resources.

Merchant_Portal

Customers use PayStream's hosted checkout page to complete payment. No integration required. 25% of transaction volume. Fee: 1.8% (higher due to support costs).

Bank_Transfer

Direct bank transfers (ACH, Faster Payments, SEPA). Lowest volume (15%) and variable fees based on routing complexity.

Merchant Types

SME (Small-to-Medium Enterprise)

Annual revenue £500K-£10M. Typically 5-50 employees. Tight cash flow. Most price-sensitive. Represent 25% of transaction volume but highest churn risk if service disrupted.

Mid-Market

Annual revenue £10M-£100M. 50-500 employees. Professional teams with financial oversight. 55% of transaction volume. Lowest churn risk.

Enterprise

Annual revenue >£100M. Large merchant operations. 20% of transaction volume. Strongest relationships but most stringent SLA requirements.

Validation Rules

Rule A: Merchant Account Status

Checks if merchant account is active and not blacklisted. Verifies account_status = "ACTIVE" AND not in blacklist table. Prevents payments from suspended or defrauded merchants.

Rule B: Transaction Amount Limits

Verifies transaction amount is within merchant's daily and monthly caps. Checks:
transaction_amount <= daily_limit AND cumulative_daily_amount <= daily_limit AND cumulative_monthly_amount <= monthly_limit. Prevents overpayment and fraud.

Rule C: Rate Limiting / Fraud Detection

Detects suspicious transaction frequency and patterns. Flags if merchant has >N transactions

per minute OR same amount charged multiple times OR transactions from multiple geographies within short timeframe. N = configurable per merchant.

Rule D: Billing Validation Rule (BVR) ⚠

Checks merchant funding and billing information before processing API_Direct payments.

Requires:

- $\text{account_balance} > \text{transaction_amount} \times 1.5$ (150% reserve requirement)
- $\text{billing_address} \text{ NOT NULL}$
- $\text{failed_transactions_last_7_days} < 5$

If ANY check fails → BILLING_VALIDATION_FAILED error returned to merchant.

Metrics & Measurements

Transaction Count

Total number of payment transactions processed in a given period (day, week, month). Unit: count.

Failed Transactions

Number of transactions that did not complete successfully (any rule failure, timeout, or merchant cancellation). Unit: count.

Failure Rate / Failure Percentage

Ratio of failed transactions to total transactions. Formula: $\left(\frac{\text{failed_transactions}}{\text{transaction_count}} \right) \times 100$. Unit: percentage (%).

Baseline Failure Rate

Historical failure rate under normal operating conditions (Weeks 1-8 for PayStream: 5.8%). Used as benchmark to detect anomalies.

Crisis Failure Rate

Elevated failure rate during problem period (Weeks 9-12 for PayStream: 12.1%). Indicates operational issue.

Root Cause Percentage

Percentage of all failures attributable to a single cause. Formula: $\left(\frac{\text{failures_from_cause}}{\text{total_failures}} \right) \times 100$. For PayStream BVR: 43.2% of all failures.

Business Metrics

Weekly Revenue Impact

Cost of lost merchant confidence due to transaction failures. Calculated as:

$\text{failed_transactions} \times \text{average_fee} \times \text{conversion_risk}$. For PayStream: ~£49,000/week.

Quarterly Impact at Risk

Estimated financial loss if crisis persists for entire quarter (4 weeks). For PayStream: £196,000 (~4 weeks × £49K/week).

Merchant Churn Risk

Number of key merchant partners threatening to leave due to service disruption. For PayStream: 8–10 merchants (representing ~£180K quarterly revenue).

SLA Violation

Failure rate exceeds contracted threshold (typically 2–5% depending on tier). PayStream SLA: <2% failure rate. Crisis at 12.1% = severe SLA violation.

Regulatory & Compliance Terms

FCA (Financial Conduct Authority)

UK financial regulator. Requires payment institutions to report transaction accuracy and any failures >40% attributable to a single cause. Notification required within 10 business days.

Payment Institution (PI)

Legal classification for companies processing payments. PayStream is FCA-authorized as PI000012345.

Transaction Accuracy Reporting

Monthly filing to FCA documenting failure rates, causes, and remediation. Required by 15th of following month (e.g., March failures → 15 April filing).

Regulatory Escalation

If root cause of failures is not documented and remediated within filing deadline, FCA may impose restrictions on PayStream's license (e.g., reduced transaction limits, operational review, fines).

Technical Terms (Simple Explanations)

Deployment

Release of code changes to production environment where real merchants' transactions are processed. For PayStream BVR: deployed 17 February 2026.

Rollout

Gradual release of changes to avoid overloading systems. PayStream BVR rollout: 10% traffic (10% of merchants) → 50% → 100% over 24 hours.

Staging

Testing environment that mimics production but uses test data. Changes tested here before production deployment.

Rollback

Reverting a deployment (undoing the change). For BVR: revert = restore previous validation rule logic. Cost: £0, Timeline: 1 hour.

Row Count

Number of data records. Data quality check: do your SQL query results match the actual database row count? If not, data may be missing or duplicated.

Simulation-Specific Terminology

Weeks 1-8 (Baseline Period)

Normal operating conditions before any deployment. Failure rate ~5.8%. Used as comparison benchmark.

Weeks 9-12 (Crisis Period)

Period after BVR deployment when failures spike. Failure rate ~12.1%. Root cause investigation period.

Week 8 Deployment

17 February 2026, 10:30 AM UTC. Billing Validation Rule modification (ticket CR-2987) deployed to production. Rollout completed 18 Feb 2:00 PM.

Crisis Deadline

4 April 2026, end of business. Hard stop for Board review of Q1 performance and remediation decision. Failure to resolve by deadline triggers Q1 miss and potential merchant churn.

FCA Filing Deadline

15 April 2026. Last day to file transaction accuracy report to Financial Conduct Authority. Must include documented root cause and remediation plan.

Key Figures (Reference)

- **Baseline Failure Rate:** 5.8% (Weeks 1-8)
- **Crisis Failure Rate:** 12.1% (Weeks 9-12)
- **API_Direct Failure Rate (Crisis):** 37% (vs. 5.2% baseline)
- **SME Impact Ratio:** 7x more affected than Mid-Market
- **Root Cause (BVR) %:** 43.2% of all failures
- **Weekly Cost:** £49,000
- **Quarterly At Risk:** £196,000
- **Merchants at Churn Risk:** 8-10 partners
- **Transaction Volume:** £1.2bn annually
- **Active Merchants:** 4,200+
- **Team Size:** 89 employees

Last Updated: March 2026

Used in Simulation: PayStream Data Analyst Crisis (6-phase, junior level)